

Statistical Methods I: Linear Model

Lec 05: Diagnostics In Simple Linear Regression

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Section 1

Intro

Review: Lecture 01-04

- Lecture 01: LSE:

$$\{\hat{\beta}_0, \hat{\beta}_1\} = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- The LSEs are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{XX}} y_i.$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \sum_{i=1}^n \left(\frac{1}{n} - \frac{\bar{x}}{S_{XX}} (x_i - \bar{x}) \right) y_i$$

- Lecture 02: CI, and hypothesis testing related to β_1 .
- Lecture 03: Estimation and prediction.
- Lecture (04): ANOVA.

Section 2

Model Diagnostics in SLR

Assumptions in SLR

- The SLR model is based on several assumptions:
 - 1 Linearity: The relationship between x and y is linear.
 - 2 Independence: The errors ϵ_i are independent. A weaker assumption is that the errors are uncorrelated.
 - 3 Homoscedasticity: The errors ϵ_i have constant variance σ^2 .
 - 4 Normality: The errors ϵ_i are normally distributed.

Model Diagnostics in SLR

- Violating any of these assumptions can lead to problems with estimation, prediction, and inference.
- Visualization is key to check these assumptions, especially plots based on residuals.

Section 3

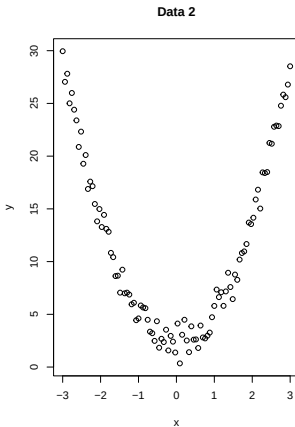
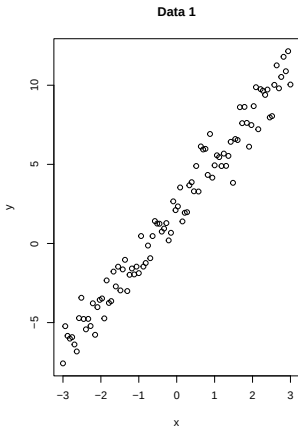
Assumption 1: Linearity

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Assumption 1 states that the relationship between x and y is linear.

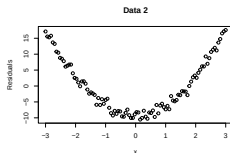
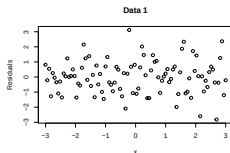
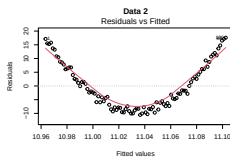
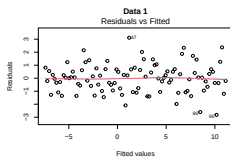
- We can check this assumption using scatter plots.
- Consequences of violating this assumption:
 - Biased estimate of regression coefficients.
 - Inaccurate predictions.
 - Invalid inference.
- How to detect? Visualization is key!

Scatter Plot of $y \sim x$



Scatter Plot based on residuals

- Residuals vs Fitted Values Plot
- Residuals vs Predictor Plot



Section 4

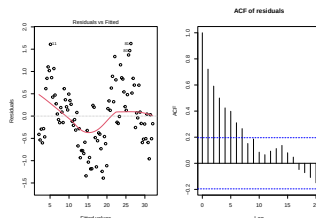
Assumption 2: Independence

Assumption 2: Independence

- Assumption 2 states that the errors ϵ_i are independent. A weaker assumption is that the errors are uncorrelated.
- This is an important assumption but not easy to check.
- When there is certain natural sequence (such as time or space) in the data we collected, the error terms can be correlated.
 - If we ignore such correlation, we may fail to utilize the information contained in the data set.

Assumption 2: Autocorrelation of residuals

- Residual plot can be used to uncover the potential correlation between errors. A more formal way to examine the assumption is to use so-called autocorrelation plot.



- In this case, we can use time series or spatial models to account for the correlation.

Section 5

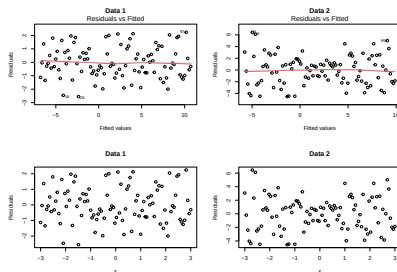
Assumption 3: Homoscedasticity

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- Assumption 3 states that the errors ϵ_i have constant variance σ^2 .
- Consequences of violating this assumption:
 - Inefficient estimates of regression coefficients.
 - Inaccurate predictions.
 - Invalid inference.

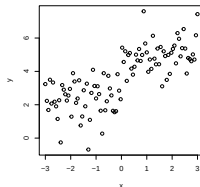
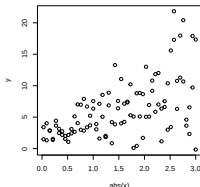
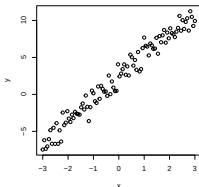
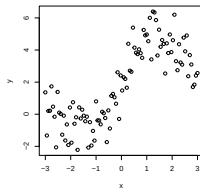
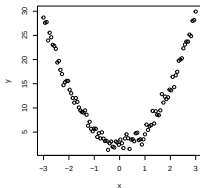
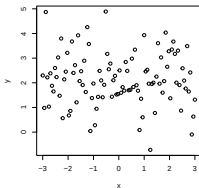
Assumption 3: Heteroscedasticity

- How to detect? Visualization.
- Constant variance means that the spread of the y (also residuals) should be roughly constant across all levels of fitted values or predictor variable.
- y vs x or fitted values
- Residuals vs x or fitted values



More examples

- In the following plots, which ones violate a linear relationship between x and y ?



Section 6

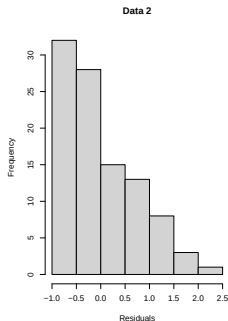
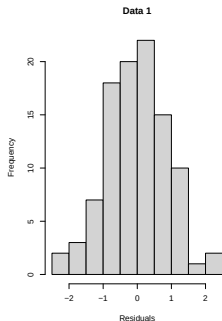
Assumption 4: Normality

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- The errors ϵ_i are normally distributed.
- Consequences of violating this assumption: if the errors do not follow a normal distribution, as long as the other assumptions holds, the consequences are usually not serious.

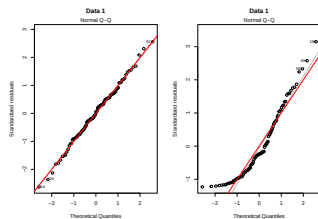
Assumption 4: Normality

- How to detect? Visualization.
- Histogram of residuals

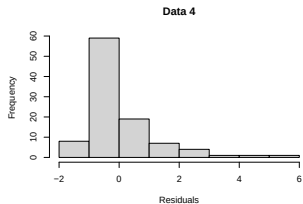
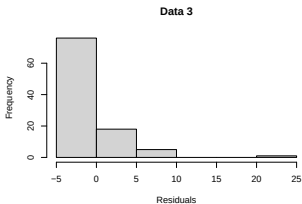
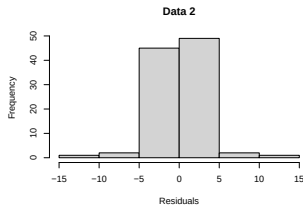
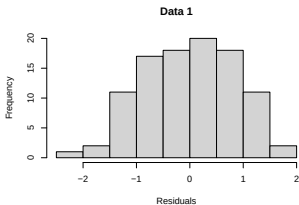


Assumption 4: Normality

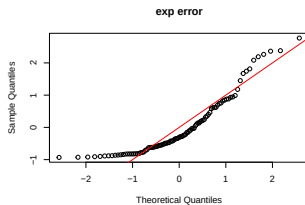
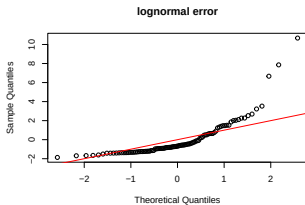
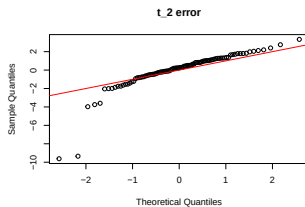
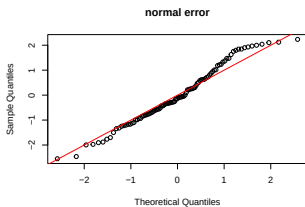
- Norm Q-Q Plot of residuals
- If the plot shows the points falling approximately along the diagonal line, then the normality assumption is satisfied.
- If the plot shows a pattern (e.g., a curve), then the normality assumption is not satisfied.



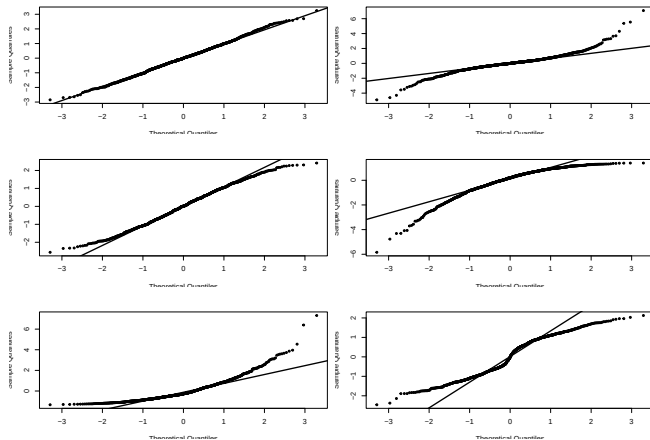
More examples of Violations of Normality



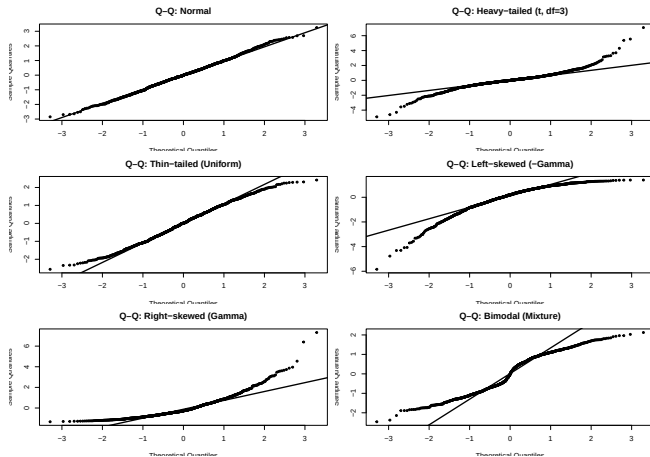
More examples of distributions: QQnorm



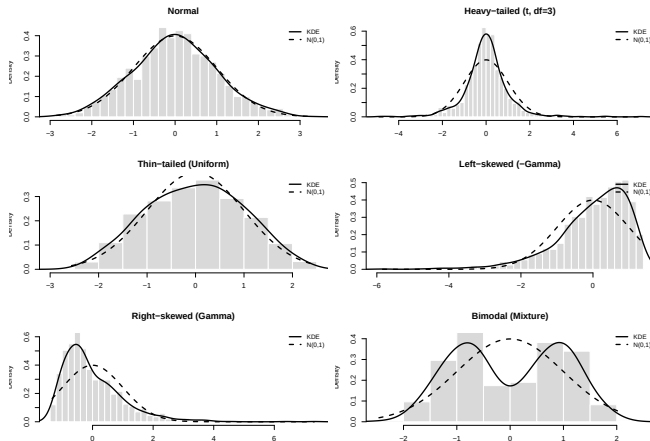
More examples of distributions: QQnorm



More examples of distributions: QQnorm



More examples of distributions: histograms



Section 7

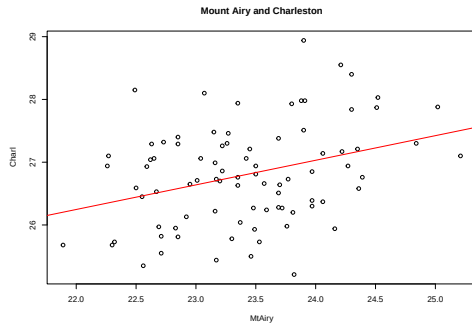
mtair

Example: mtair

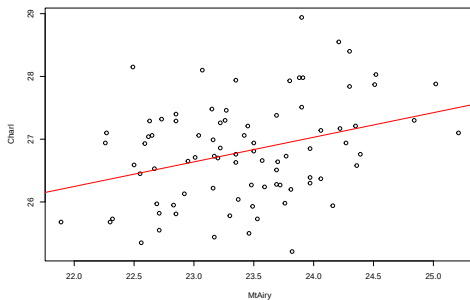
Summer temperature averages in Mount Airy, NC and Charleston, SC (1937-2021)

```
mtair=read.csv("data/mta1.csv", head=TRUE)
plot(mtair$MtAiry, mtair$Charl,
     xlab="MtAiry",
     ylab="Charl",
     main="Mount Airy and Charleston")
fit <- lm(Charl ~ MtAiry, data = mtair)
abline(fit, col="red", lwd=2)
```

Example: mtair

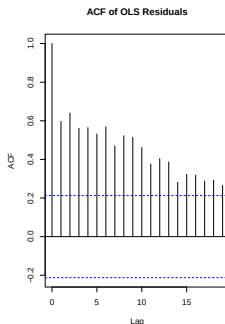
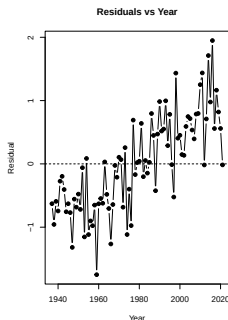


mtair: Assumption 1: Linearity



mtair: Check for Assumption 2 (Independence)

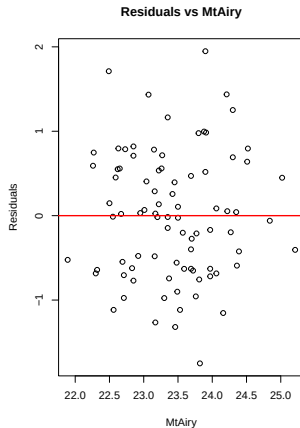
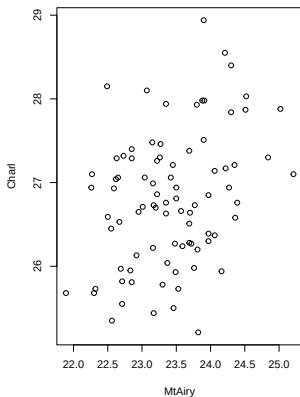
- Recall that the observations are ordered by year (1937-2021), in the chronological order.



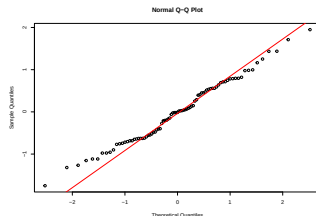
mtair: Check for Assumption 2 (Independence)

- There are also tests for serial correlation, such as
 - Durbin-Watson test
 - Breusch-Godfrey test
 - Ljung-Box test

mtair: Check for Assumption 3 (Homoscedasticity)



mtair: Check for Assumption 4



Section 8

Battery Life

Example: Battery data

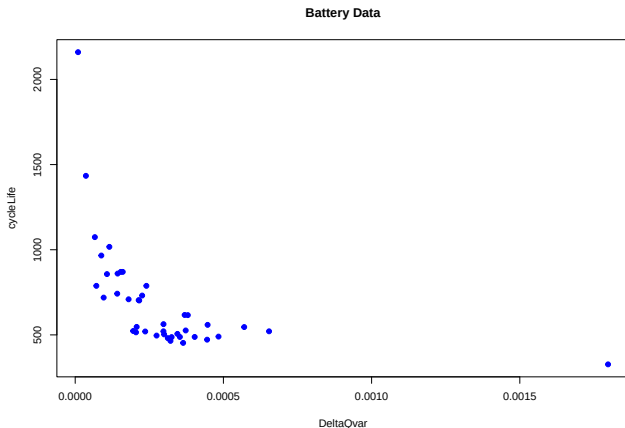
Background: Predicting cycle life of lithium-ion batteries from initial cycles

Source: MathWorks Predictive Maintenance Toolbox

- x : difference in discharge capacity as a function of voltage between 100th and 10th cycle
- y : cycle life (number of cycles until the battery capacity falls below 80% of its initial capacity)

Example: Battery data

Scatter plot:



Example: Battery data

- Distributions of x and y are highly right-skewed.

